Synthetic Cross-Lingual Data Generation for Multimodal Speaker Identification

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Abstract

This work presents a synthetic data generation pipeline for multimodal speaker identification under cross-lingual and missing-modality settings. The goal is to improve the robustness of the baseline face–voice association model by augmenting the original training data with generated cross-lingual audio-visual samples. The proposed pipeline takes an input video, extracts the speech signal, transcribes and translates it into the target language, generates translated speech while preserving speaker characteristics, synchronizes the generated audio with the speaker's mouth movements, and finally extracts audio and visual embeddings. ECAPA-TDNN is used for audio embedding extraction, while FaceNet is used for visual embedding extraction. For the visual modality, FaceNet embeddings are extracted from every 25th frame to reduce redundancy while preserving representative facial information. The final experiments compare the baseline trained on original data, synthetic data, and mixed original-plus-synthetic data using the P3–P6 evaluation protocol.

Index Terms: multimodal learning, speaker identification, cross-lingual learning, missing modality, synthetic data

1. Experimental Setup

The assignment focuses on multimodal speaker identification under two challenging conditions: cross-lingual generalization and missing modalities. In the cross-lingual setting, the model is trained using one language and evaluated on another. In the missing-modality setting, one input modality is unavailable during evaluation.

The original dataset consists of pre-extracted audio and visual embeddings. Audio embeddings are obtained using ECAPA-TDNN, while visual embeddings are obtained using FaceNet. These original embeddings are used as the baseline feature set. In addition, we create synthetic cross-lingual data from raw English videos and generate German synthetic samples. The aim is to increase speaker-level training diversity and improve robustness when the test language changes or when the visual modality is unavailable.

1.1. Synthetic Data Generation Pipeline

The proposed synthetic data pipeline consists of five main stages. First, the audio track is extracted from the input video using FFmpeg. Second, Whisper is used for automatic speech recognition and language detection. This stage converts the spoken content into text and identifies the source language. Third, the transcribed text is translated into the target language, in our case German. Fourth, Coqui XTTS is used to generate speech in the target language using the original audio as a speaker reference. This allows the generated speech to preserve speaker-specific vocal characteristics more closely than ordinary text-to-speech systems. Finally, Wav2Lip is used to synchronize the generated audio with the speaker's mouth movements in the video.

If Wav2Lip fails to detect a face in a video, the generated translated audio is still preserved, while the original video is used as the visual source for FaceNet extraction. This fallback is used because the speaker identity in the visual modality remains unchanged, and the main synthetic variation is introduced through the translated audio.

1.2. Feature Extraction

After generating synthetic samples, ECAPA-TDNN is used to extract speaker embeddings from the generated audio. For the visual modality, FaceNet embeddings are extracted from the generated or fallback video. To avoid producing thousands of highly redundant visual embeddings from nearly identical neighboring frames, we extract FaceNet features from every 25th frame. This follows the idea of keeping representative visual information while reducing computational cost and dataset imbalance.

The synthetic embeddings are saved in the same format as the original feature files. The resulting CSV files follow the same structure as the provided feature tracker files and include paths to ECAPA-TDNN embeddings, FaceNet embeddings, identity labels, and identity names. Three training sets are then considered: original data only, synthetic data only, and mixed original plus synthetic data.

2. Experiments

The experiments are designed to evaluate whether synthetic cross-lingual data improves the robustness of the baseline model under both cross-lingual and missing-modality settings. We consider three training configurations: original data only, synthetic data only, and mixed data. The original data corresponds to the provided German training embeddings. The synthetic data consists of generated German samples obtained from English source videos using the proposed translation and lip-synchronization pipeline. The mixed setting combines original German training data with synthetic German samples.

We perform two types of evaluation. First, we evaluate the trained models on the official provided test embeddings. This is the main evaluation because it is directly comparable to the baseline setting used in previous experiments. Second, we evaluate the same trained models on re-extracted English embeddings produced by our own ECAPA-TDNN and FaceNet extraction pipeline. This second evaluation is not intended to replace the official test set, but rather to verify that our feature

Table 1: *Evaluation on provided test embeddings. Accuracy is reported for P3–P6 and averaged across all settings.*

Training Data	P3	P4	P5	P6	Avg.
Orig data	89.18	76.02	93.10	77.16	83.87
Synth data	96.78	79.24	94.83	56.03	81.72
Mixed 10%	98.83	85.09	98.28	86.64	92.21

Table 2: *Evaluation on re-extracted English embeddings matched to the official English test split.*

Training Data	English FA	English Audio	Avg.
Orig data	95.97	83.83	89.9
Synth data	90.63	61.72	76.175
Mixed 10%	99.28	95.13	97.205

extraction pipeline produces embeddings compatible with the baseline model.

2.1. Evaluation on Provided Embeddings

The first evaluation uses the official feature tracker files provided with the assignment. The model is trained on German data and evaluated on both German and English test splits. The evaluation follows the P3–P6 protocol:

- **P3**: German test set with both face and audio available.
- **P4**: German test set with the face modality missing, using audio only.
- **P5**: English test set with both face and audio available.
- **P6**: English test set with the face modality missing, using audio only.

The missing-modality setting is simulated by setting the face embeddings to zero during evaluation. Classification accuracy is used as the evaluation metric.

2.2. Evaluation on Re-Extracted English Embeddings

In addition to the official evaluation, we evaluate the trained models on English embeddings extracted using our own pipeline. To make this evaluation comparable to the official English test split, the re-extracted English CSV is filtered to match the clips appearing in `v3_test_English.csv`. Matching is performed using speaker identity, video ID, and clip ID.

This evaluation allows us to test whether the implemented ECAPA-TDNN and FaceNet extraction pipeline produces embeddings that are compatible with the baseline model. Since only English videos were re-extracted, this evaluation focuses on English face-audio and English audio-only performance.

2.3. Synthetic Data Cleaning

During preliminary experiments, we observed that some synthetic samples were generated from English clips that also appeared in the official English test split. This created potential train-test leakage and resulted in unrealistically high unseen-language accuracy. To address this issue, we created a cleaned synthetic training set by removing all generated samples whose source clips overlapped with `v3_test_English.csv`. The cleaned synthetic set is used for the final comparison.

This ensures that the synthetic training data does not contain transformed versions of the English evaluation clips. Therefore, the final results better reflect cross-lingual generalization rather than memorization of test identities or clips.

3. Discussion

The results show that synthetic cross-lingual data can improve the baseline model, but its effect depends strongly on how the synthetic data is used. On the provided test embeddings, the original-only baseline achieves an average accuracy of 83.87%. Training only on the cleaned synthetic data gives a lower average of 81.72%, even though it improves P3 and P4 compared to the original baseline. This indicates that synthetic samples contain useful speaker identity information, but synthetic data alone does not fully replace the original training distribution.

The strongest result is obtained with the mixed clean 10% setting, which combines the original German training data with a leakage-free subset of synthetic German samples. This setting achieves the best average accuracy of 92.21%, outperforming both the original-only and synthetic-only settings. The improvement is especially important in P4 and P6. In P4, the audio-only in-language accuracy increases from 76.02% to 85.09%. In P6, the cross-lingual audio-only accuracy increases from 77.16% to 86.64%. These two settings are the most relevant for missing-modality robustness, because the face modality is unavailable at test time. The improvement suggests that synthetic translated audio can help the model learn more robust audio-based speaker representations.

The synthetic-only result also provides an important observation. Although it achieves high performance in the multimodal settings, especially P3, it performs poorly in P6 compared to the original and mixed settings. This means that synthetic data alone may introduce a domain gap, especially when the model is tested in the cross-lingual audio-only condition. In contrast, the mixed setting benefits from both sources: the original data preserves the real feature distribution, while the synthetic data adds additional cross-lingual variation.

The second evaluation, using re-extracted English embeddings, shows a similar trend. The mixed clean 10% model achieves the highest average accuracy of 97.21%, with 99.28% in the English face-audio setting and 95.13% in the English audio-only setting. This confirms that the implemented feature extraction pipeline produces embeddings that are compatible with the baseline model. However, this second evaluation should be interpreted as a pipeline validation rather than the main benchmark, because the embeddings were extracted using our own preprocessing and frame-sampling strategy. The provided-embedding evaluation remains the primary comparison.

A critical part of this work was preventing data leakage. In preliminary experiments, we observed unrealistically high unseen-language accuracy when training on the initial synthetic set. After checking the source clips, we found that some synthetic training samples had been generated from English videos that also appeared in the official English test split. This means that the model could indirectly see transformed versions of test clips during training. To address this, we removed all synthetic samples whose source identity, video ID, and clip ID overlapped with `v3_test_English.csv`. The final synthetic clean and mixed clean datasets therefore exclude transformed versions of English test clips. This cleaning step is necessary to ensure that improvements reflect cross-lingual generalization rather than memorization of evaluation data.

Overall, the results support the usefulness of synthetic cross-lingual augmentation, but they also show that synthetic data must be used carefully. The best performance comes from combining original and cleaned synthetic data, rather than replacing the original data entirely. This suggests that synthetic data is most effective as an augmentation strategy, especially for improving missing-modality and cross-lingual missing-modality performance.

4. References

[1] Saeed, M. S., Khan, M. H., Nawaz, S., Yousaf, M. H., and Del Bue, A. "Fusion and Orthogonal Projection for Improved Face-Voice Association." In *ICASSP 2022 - IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 7057–7061, 2022.